GRACE V. RINGLEIN

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EDUCATION

PhD Candidate in Biostatistics, Johns Hopkins Bloomberg School of Public Health — 2022–2027 (Expected)

Master of Science in Physics & Astronomy, University of Pennsylvania — 2017–2020

Bachelor of Arts in Physics, University of Pennsylvania — 2016–2020

- Minor in Fine Arts, Concentration in Computer Techniques
- Summa Cum Laude; Phi Beta Kappa; with Honors Distinction in Physics
- Graduate of the Roy and Diana Vagelos Program for the Molecular Life Sciences

RESEARCH EXPERIENCE

JOHNS HOPKINS BLOOMBERG SCHOOL OF PUBLIC HEALTH

Doctoral Research, Department of Biostatistics — Dec. 2022–Present
Advisors: Elizabeth Stuart, PhD; Trang Nguyen, PhD
Key Collaborators: Harsh Parikh, PhD (Yale University School of Public Health); Peter Zandi, PhD (Johns Hopkins School of Medicine)

- Dissertation topic: Causal inference under measurement error and unobserved confounding: methods and applications in electronic health records research.

Collaborative Research, Department of Epidemiology — Aug. 2025–Present
Principal Investigator: John Jackson, ScD

- Participating in a cross-disciplinary systematic scoping review of conceptual models for measuring disparity in medicine and public health.
- First author on working paper focused on conceptual models based on comparison of "similarly situated" individuals or groups.

Collaborative Research, Department of Health Policy and Management — Dec. 2022–Present
Principal Investigator: Catherine Ettman, PhD

- Analysis and ongoing statistical support for investigations of the effect of job loss on mental health (and the potential moderating role of financial strain) using national survey data.

Collaborative Research, Precision Medicine Center of Excellence in Mood Disorders — Jun. 2023–Present
Principal Investigators: Peter Zandi, PhD; Elizabeth Stuart, PhD; Catherine Ettman, PhD

- Cohort construction and power analyses supporting Dr. Ettman's K01 grant *Economic Precarity, Costs, and Continuity of Mental Health Care* (now funded).
- Analysis and ongoing statistical support for research projects related to access and continuity of mental health care using electronic health records.
- Statistical support for projects related to comparative effectiveness of antidepressants.

NATIONAL INSTITUTE OF MENTAL HEALTH

Post-baccalaureate Fellow (IRTA), Section on Development and Affective Neuroscience — 2020–2022
Principal Investigator: Daniel Pine, MD

- Co-authored publications related to pediatric mental health using neuroimaging data.
- Conducted early data analysis and analytic planning for projects applying large language models to classify behaviors in therapy session transcripts.

UNIVERSITY OF PENNSYLVANIA

Undergraduate Research, Multiple laboratories 2017–2020

- Irina Marinov, PhD, Department of Earth & Environmental Science: Investigated convection dynamics in Southern Ocean climate models (May 2019–May 2020)
- Emily Falk, PhD, Annenberg School of Communication: Created novel stimuli for MRI studies of addiction. (November 2019–May 2020).
- Masao Sako, PhD, Department of Physics & Astronomy: Constructed a pipeline for detecting microlensing events in Dark Energy Survey data. (June 2017–December 2018).

TEACHING EXPERIENCE

AMERICAN UNIVERSITY OF ARMENIA

Visiting Lecturer, Turpanjian College of Health Sciences Oct 26–Nov 4, 2026

- *Inferential Biostatistics* (PH 321)
  - (Upcoming): delivering daily lectures and lab exercises in Stata; for 15-20 students.

JOHNS HOPKINS BLOOMBERG SCHOOL OF PUBLIC HEALTH

Teaching Academy Certificate Program, 2025–Present
Johns Hopkins University Center for Teaching Excellence and Innovation

- Developing "refresher" videos on statistical topics where students often struggle.

Teaching Assistant, Department of Biostatistics 2023–Present

- *Statistical Methods in Public Health I, II, and IV* (PH 140.621-622, and 624, respectively)
  - **Lead TA** (PH 140.621, 1st term, 2026): currently teaching multiple 90-minute lab sessions per week, reviewing course topics and introducing coding fundamentals in R.
  - **TA** (PH 140.621-622, 1st and 2nd term, 2024): held weekly office hours with co-TAs; graded 40-50 students per assignment, 4 assignments per term.
  - **TA** (PH 140.624, 4th term, 2026): held five office hours per week.
- *Causal Inference in Medicine and Public Health I* (PH 140.664)
  - **TA** (PH 140.664, 3rd & 4th terms, 2024; 3rd & 4th terms, 2025; 3rd term, 2026)
    - Led weekly tutorials on applying causal inference methods using R (8 tutorials per term in 2024, 4 per term in 2025, 2026); updating and developing materials; recorded three tutorials for ongoing use.
    - Guest lectures on Regression Discontinuity Designs (Term 3, 2025; Term 3, 2026).
    - Graded essay questions on assignments (50+ students; 1-2 assignments per term).
- *Multilevel and Longitudinal Models* (PH 140.655-656)
  - **TA** (PH 140.655, 1st term, 2025): held rotating office hours with co-TAs; graded assignments (60+ students, 2 assignments).
  - **TA** (PH 140.656, 2nd term, 2025): held weekly office hour focused on student projects; graded one quiz and one project (29 students).
- *Advanced Methods in Biostatistics I–II* (PH 140.751-752)
  - **TA** (PH 140.751-752, 1st and 2nd terms, 2023): Sole TA; held weekly office hour; graded (4 assignments per term; 12 students).

Tutoring, Department of Biostatistics 2025–2026
Individual tutoring for a Biostatistics PhD student preparing for comprehensive exams

TA Training Planning Committee, Department of Biostatistics 2025, 2026
Assisted with preparation and delivery of half-day training for new Biostatistics TAs.

UNIVERSITY OF PENNSYLVANIA

Teaching Assistant, Department of Physics and Astronomy 2018–2020

- *Laboratory Electronics* (PHYS 364)
  - **TA** (Spring 2020): attended laboratories (three hours, twice a week) to troubleshoot and answer conceptual questions; graded lab reports.
- *Physics for Architects* (PHYS 008)
  - **TA** (Fall 2019): Hosted a bi-weekly homework help session, for a general undergraduate audience with varied mathematical background; graded problem sets.
- *General Physics: Electricity, Magnetism, Optics, Modern Physics* (PHYS 102 Lab)
  - **Laboratory Instructor** (Fall 2019; Spring 2018): delivered brief lectures to introduce weekly lab topics; guided students (primarily pre-medical undergraduates) through experiments and provided conceptual and technical support; graded lab reports.

HONORS AND AWARDS

Society for Causal Inference Travel Award, American Causal Inference Conference 2026

Lodging Award, Statistics in the Age of AI Conference, George Washington University 2024

Best Poster Award, Society for Epidemiologic Research Annual Meeting 2024
- "Trends in the Use of Telehealth by Area Deprivation in a Large U.S. Academic Hospital" (presented by Catherine Ettman).

Roy and Diana Vagelos Science Challenge Award, University of Pennsylvania 2019

NASA Pennsylvania Space Grant Consortium Scholarship, PA Space Grant Consortium 2018

Francis A. Velay Fellowship, University of Pennsylvania 2017

COMMUNITY, LEADERSHIP, & SERVICE

Biostatistics Student Organization, Johns Hopkins Bloomberg School of Public Health 2023–2026
- Mentorship Committee Co-lead (2025–26)
- Curriculum Committee Representative for the PhD Program (2023–24, 2024–25)
- Mental Health Graduate Network Representative for Biostatistics (2023–24, 2024–25)
- Peer mentor for one first-year Biostatistics PhD student (2023–24, 2024–25)

Alumni Mentor, Pennsylvania Governor's School for the Sciences 2022–2023

Planetarium Presenter, Maryland Science Center Summer 2022

Community Arts Instructor, Baltimore ClayWorks Summer 2022

Research Assistant Onboarding Co-lead, Section on Development & Affective Neuroscience Summer 2021

Vagelos Happiness Project Mentor, University of Pennsylvania 2019–2020

Peer Advisor, University of Pennsylvania 2019–2020

PUBLICATIONS

PREPRINTS, UNDER REVIEW, OR IN PREPARATION

1. **Ringlein, G. V.**, Nguyen, T. Q., Zandi, P. P., Stuart, E. A., & Parikh, H. Demystifying proximal causal inference. Revising for resubmission to *Statistical Science*. doi:10.48550/arxiv.2512.24413.

2. **Ringlein, G. V.**, Nguyen, T. Q., Stuart, E. A., & Parikh, H. (2026). Proximal causal inference through cross-proxy balancing. Submitted. doi:10.48550/arXiv.2609.38175
3. **Ringlein, G. V.**, Sun, X., Parikh, H., Ferryman, K., & Jackson, J. Similarly situated conceptual models for disparity and fairness. In preparation.

PEER-REVIEWED PUBLICATIONS

1. Nguyen, T. Q., Nabi, R., Yang, F., **Ringlein, G. V.**, & Stuart, E. A. Self-separated and self-connected models for mediator and outcome missingness in mediation analysis. *Statistical Science*. In press. https://arxiv.org/abs/2411.07221
2. Ettman, C. K., Jeon, K., Dohlman, P., **Ringlein, G. V.**, Straub, J., et al. Psychiatric services following suicide-related hospital visits: An analysis of electronic health records among patients with depression. *Psychiatric Research and Clinical Practice*. In press.
3. Mady, L. J., **Ringlein, G. V.**, Satpathy-Horton, R., Stuart, E. A., Galea, S., Ettman, C. K. (2026). Pre-existing financial circumstance, job loss, and probable depression among U.S. adults: a prospective population-based cohort study. *Lancet Regional Health - Americas*, 63. doi:10.1016/j.lana.2026.101590
4. Hong, H., Zhao, C., Jeon, K., Goicoechea, E. B., Brantner, C. L., **Ringlein, G.**, et al. (2026). Comparative effectiveness of antidepressants for depression using EHRs from two health systems. *BMC Psychiatry* 26, 638. doi:10.1186/s12888-026-08159-x
5. Richier, C., Zugman, A., Harrewijn, A., Cardinale, E., Khosravi, P., ..., **Ringlein, G. V.**, et al. Brain Age Prediction in Generalized Anxiety Disorder using a Convolutional Neural Network. *Translational Psychiatry* 16, 368 (2026). doi:10.1038/s41398-026-04078-3
6. Ettman, C. K., **Ringlein, G. V.**, Dohlman, P., Straub, J., Brantner, C. L., Chin, E. T., et al. (2025). Trends in mental health care and telehealth use across area deprivation: An analysis of electronic health records from 2016–2024. *PNAS Nexus*, pgaf016. doi:10.1093/pnasnexus/pgaf016
7. Ettman, C. K., Brantner, C. L., Goicoechea, E. B., Dohlman, P., **Ringlein, G. V.**, Straub, J., et al. (2025). Gaps in psychiatric care before and after the COVID-19 pandemic among patients with depression using electronic health records. *Psychiatry Research*, 344, 116354.
8. Brantner, C. L., Yu, W., Zhao, C., Jeon, K., **Ringlein, G. V.**, Wang, Q., et al. (2025). The challenges of integrating diverse data sources: A case study in major depression. *Health Services and Outcomes Research Methodology*. doi:10.1007/s10742-025-00363-8
9. Lossio-Ventura, J. A., Frank, S., **Ringlein, G.**, Bonson, K., Olszko, A., Knobel, A., et al. (2025). Automated classification of exposure and encourage events in speech data from pediatric OCD treatment. *JAMIA Open*, 8(6), ooaf127. doi:10.1093/jamiaopen/ooaf127
10. Zugman, A., **Ringlein, G. V.**, Finn, E. S., Lewis, K. M., Berman, E., Silverman, W. K., et al. (2025). Brain functional connectivity and anatomical features as predictors of cognitive behavioral therapy outcome for anxiety in youths. *Psychological Medicine*, 55, e91. doi:10.1017/S0033291724003131
11. Kirk, P. A., Qamar, P., Zugman, A., Abend, R., Frank, S., **Ringlein, G. V.**, et al. (2025). The relations among anxiety, movie-watching, and in-scanner motion. *Human Brain Mapping*, 46(4), e70163. doi:10.1002/hbm.70163
12. Medel, V., Alliende, L. M., Bethlehem, R., Seidlitz, J., **Ringlein, G.**, Arango, C., et al. (2025). Human development, inequality, and their associations with brain structure across 29 countries. *European Psychiatry*, 68(1), e100. doi:10.1192/j.eurpsy.2025.10060
13. **Ringlein, G. V.***, Ettman, C. K.,* & Stuart, E. A. (2024). Income or job loss and psychological distress during the COVID-19 pandemic. *JAMA Network Open*, 7(7), e2424601. doi:10.1001/jamanetworkopen.2024.24601
    - *Co-first authors
14. Bruin, W. B., Zhutovsky, P., van Wingen G., Bas-Hoogendam., J. M., Groenewold, N. A., et al. Brain-Based Classification of Youth with Anxiety Disorders: an ENIGMA-ANXIETY Transdiagnostic Examination using Machine Learning. *Nature Mental Health* 2, 104–118 (2024). doi:10.1038/s44220-023-00173-2

15. Zugman, A., Alliende, L. M., Medel, V., Bethlehem, R. A. I., Seidlitz, J., **Ringlein, G.**, et al. (2023). Country-level gender inequality is associated with structural differences in the brains of women and men. *Proceedings of the National Academy of Sciences*, 120(20), e2218782120. doi:10.1073/pnas.2218782120
16. Ettman, C. K., Fan, A. Y., Philips, A. P., Adam, G. P., **Ringlein, G.**, Clark, M. A., et al. (2023). Financial strain and depression in the U.S.: A scoping review. *Translational Psychiatry*, 13(1), 168. doi:10.1038/s41398-023-02460-z
17. Harrewijn, A., Cardinale, E. M., Groenewold, N. A., Bas-Hoogendam, J. M., Aghajani, M., Hilbert, K., et al. (2021). Cortical and subcortical brain structure in generalized anxiety disorder: Findings from 28 research sites in the ENIGMA-Anxiety Working Group. *Translational Psychiatry*, 11(1), 502. doi:10.1038/s41398-021-01622-1
18. Gnanadesikan, A., Speller, C. M., **Ringlein, G.**, San Soucie, J., Thomas, J., & Pradal, M.-A. (2020). Feedbacks driving interdecadal variability in southern ocean convection in climate models: A coupled oscillator mechanism. *Journal of Physical Oceanography*, 50(8), 2227–2249. doi:10.1175/JPO-D-20-0037.1

PRESENTATIONS

1. **Ringlein, G. V.**, Nguyen, T. Q., Zandi, P., Stuart, E. A., & Parikh, H. Demystifying Proximal Causal Inference.
   - *Joint Statistical Meetings*, 2026. Boston, MA. Oral Invited Session.
   - *ENAR Spring Meeting*, 2026. Indianapolis, IN. Oral Invited Session.
   - *American Causal Inference Conference*, 2026. Salt Lake City, UT. Contributed Session.
2. **Ringlein, G. V.**, Parikh, H., Nguyen, T. Q., & Stuart, E. A., Comparing Proxy Variable Methods for Reducing Omitted Variable Bias in EHR-Based Causal Inference. *Joint Statistical Meetings*, 2025. Nashville, TN. Contributed Poster Presentation.
3. Ettman, C. K., **Ringlein, G. V.** (presenter), Straub, J., Dohlman, P., Chin, E.T., ..., Stuart, E.A., & Zandi, P.P. What are the appointment and patient predictors of no-show in a post-pandemic world? *Academy Health Annual Research Meeting*, 2024. Baltimore, MD. Contributed Poster Presentation.
4. **Ringlein, G. V.** and Chin, E. T. Impacts of Predictive Uncertainty in Healthcare Prior Authorization. *Statistics in the Age of AI Conference*, 2024. George Washington University. Contributed Poster Presentation.
5. **Ringlein, G. V.,** Ettman, C. K., and Stuart, E. A. Power Analyses for Regression Discontinuity Designs with Binary Outcomes. *ENAR Spring Meeting*, 2024. Baltimore, MD. Contributed Poster Presentation.
6. **Ringlein, G. V.**, Winkler, A., Pine, D., & Zugman, A. Effects of preprocessing on connectome fingerprinting in a sample of anxious and healthy youth. *Society for Neuroscience Annual Meeting*, 2021. Virtual. Contributed Poster Presentation.

TECHNICAL SKILLS

- **Programming:** R, Python, Stata, Bash, MATLAB, SQL, Java
- **Graphics:** Adobe Photoshop, Adobe Illustrator

REFERENCES

- Elizabeth Stuart, PhD. Professor and Chair of the Department of Biostatistics, Johns Hopkins Bloomberg School of Public Health. Email: estuart@jhu.edu.
- Peter Zandi, PhD. Professor of Psychiatry and Behavioral Sciences, Johns Hopkins School of Medicine. Email: pzandi1@jhu.edu.
- Harsh Parikh, PhD. Assistant Professor of Biostatistics, Yale University School of Public Health. Email: harsh081193@gmail.com.